Bundling or Silence: A Pre-Registered Benchmark of Multi-Issue Nash Bargaining in Mixed-Ownership Robot Fleets

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ABSTRACT

Twenty-four robots owned by different parties mine ore and haul it to a sink in a deterministic gridworld, under battery budgets, lossy energy transfer, and a fixed horizon. Robots can strike bundled deals trading energy, cargo, and territory claims. Owners differ, so every deal must be individually rational (IR): no side accepts a loss. We benchmark Nash-bargained bundles against altruistic rescue, bilateral and broadcast sequential single-item (SSI) auctions, and a cooperative ceiling, under a mean-preserving heterogeneity dial. Three claims were registered before results. C1: single-issue IR trade is structurally infeasible — only bundles trade. C2: bargaining with a rescue floor out-delivers the bilateral auction on two geometries; against the strong SSI baseline a registered kill fired — bargaining buys survival, not throughput. C3: the bargaining–auction gap grows with heterogeneity, then saturates. On moving fields, auction coverage out-explores bargaining; a map-information market improves audit integrity, not output. Our first registration failed hostile review; every fired kill is reported.

KEYWORDS

multi-robot systems, market-based coordination, multi-issue negotiation, Nash bargaining solution, pre-registration, benchmark

1 INTRODUCTION

Robot deployments increasingly cross ownership boundaries: delivery fleets in facilities owned by others, robots from different vendors sharing chargers, mixed fleets in one warehouse. This is commercial practice, not a hypothetical — robots-as-a-service contracts already price physical work per outcome (AutoStore bills warehouse picks per pick,[1] and GXO pays Agility Robotics for humanoid labor by utilization[2]) — so a coordination step between robots with different owners is also a commercial event between firms. The standard premise of multi-robot coordination — one shared objective — is then false by construction. Each robot answers to its owner, and any step that leaves an owner worse off will eventually be disabled. Coordination must be *individually rational* (IR): no participant accepts a loss.

The market-based multi-robot lineage [10, 13, 14] saw this early and imported the auction: tasks are announced, robots bid, the cheapest capable robot wins. But a bid is one scalar. It cannot express the deal mixed-ownership logistics needs — "I take your two crates because I am efficient and you are nearly dead; you top up my charge and cede your claim on the near source." Multi-issue negotiation (logrolling) is mature in the automated negotiation community [3, 5], yet an adversarially verified literature sweep (Section 2) found no published system in which robots strike bundled multi-issue agreements with each other.

This paper contributes a benchmark, not a deployment: a controlled world in which the bargaining layer's value can be isolated, measured, and killed. Three registered questions: is bundling *necessary* for IR trade (C1)? How much of the cooperative ceiling does bundled Nash bargaining recover — the residual being the *coordination gap* (C2)? Does the advantage over single-issue auctions grow with heterogeneity (C3)?

Two objections deserve pre-emption. First, with known utilities the Nash bargaining solution is a computation, not a protocol; the benchmark therefore treats information quality as a variable, and under estimation error and lies (v5–v7) the mechanism's measured value is *deal integrity* — the true-loss veto turns error into failed proposals — not cleverness. Second, our agents are simulated too; what separates them from the negotiation literature's is not hardware but where utilities come from. Issue values are computed by executing the candidate deal in the world's dynamics — movement costs, the 25% transfer loss, time-consuming exchanges, stranding — not drawn from exogenous preference tables; hence the evaluated-Φ==executed-Φ assertion on every deal. We call this *physically grounded.*

Positioning: this is market-based multi-robot / MAS work, not swarm robotics — N=24, full-information utilities, and a deterministic gridworld fail the field's own criteria for "swarm" [28], and we use the word only to say so. All claims were pre-registered with explicit kill conditions before results — including a v1 registration that failed hostile review and was discarded (Section 3.4) — and fired kills are reported as results, not buried.

2 RELATED WORK

This section follows an adversarially verified prior-art map (20 primary sources fetched; 25 claims checked by 3-vote panels: 23 confirmed 3-0, 2 refuted; `LITERATURE.md` in the code release). Its verdict: the intersection of multi-issue negotiation and inter-robot coordination is unoccupied.

Market-based multi-robot coordination is single-issue. MURDOCH [13] assigns tasks by first-price one-round auctions; a bid is one scalar fitness value. TraderBots [33] extends auctions to task trees, but a bid remains task + price + tree — a full-text search of the thesis (187 pp.) finds no Nash, logrolling, multi-issue, bargaining, or energy exchange. The MRTA taxonomy [14] and the Dias et al. survey [10] frame the field's mechanisms; none bundle issues. Later robotics "negotiation" stays single-issue: task

[1] `autostoresystem.com/insights/buying-vs-raas-whats-the-best-strategy-for-investing-in-warehouse-robotics`, accessed 2026-07.
[2] `investors.gxo.com/news-releases/news-release-details/gxo-signs-industry-first-multi-year-agreement-agility-robotics`, accessed 2026-07.

reallocation [9]; opponent-utility modeling for task allocation [20].

Multi-issue IR contracting among self-interested agents is not new. Sandholm's contract types [29] give self-interested agents *cluster* (bundle) contracts, *swaps*, and multi-agent contracts with side payments under IR; each escapes local optima the others cannot, and atomic OCSM contracts reach the global optimum in theory. Andersson and Sandholm [1] sequence contract types for anytime reallocation. Multiagent resource allocation (MARA) covers the same ground over abstract goods: Chevaleyre et al. [8] survey allocation by negotiation; Endriss et al. [11] characterize the deal classes — including multilateral bundle deals — needed for socially optimal allocations under IR. Our absence claim is therefore narrow: missing is a *physically grounded* instantiation with *heterogeneous physical issue types* — lossy energy vs cargo vs claim rights, each with its own dynamics — under *executed dynamics*, benchmarked empirically rather than proved as reachability results. That gap is the one this benchmark occupies.

Two terminology traps. Combinatorial bids are not multi-issue deals: Lin and Zheng [23] auction bundles of *tasks* at one scalar price per bundle — one issue type, no cross-issue trade-off. *Nash equilibrium is not Nash bargaining*: a 2025 EAAI paper [17] applies Nash *equilibrium* analysis to a non-cooperative task-selection game — no offers, no deals, no exchange; energy sits inside each robot's own utility. From the other direction, the 2026 line applying the Nash bargaining solution to vehicle-to-vehicle energy markets (e.g., arXiv:2605.22363) bargains one issue — price/quantity of energy — over purely economic utilities. Keyword searches ("combinatorial," "Nash, robots, energy") wrongly suggest the niche is occupied.

Robot-robot energy exchange exists in hardware, but is rule-based. The CISSBot line demonstrates physical battery swapping [30] under Ngo and Schiøler's probabilistic "randomized trophallaxis" [27] — no offer or acceptance step. Moonjaita et al. [25] fix roles by battery thresholds and amounts by energy averaging: numerically an egalitarian split, no agreement struck. Virtual trophallaxis [31] couples energy to navigation as signaling. The physical channel our benchmark abstracts exists; the bargaining layer above it does not.

The negotiation community never crossed into robot-robot coordination. ANAC ran 2010–2015 entirely in the Genius simulator over table-defined preference profiles; the organizers' retrospective [5] contains zero occurrences of robot, embodied, or physical, and the 2015 roadmap targeted marketplaces, energy markets, telecom. ANAC 2025 [3] remains virtual, with LLM integration as the forward direction. The only robotics crossover is dyadic human-robot *social* negotiation [4] — not an inter-robot primitive.

LLM × multi-robot work is single-dimension dialogue. MARLIN [15] negotiates the next joint movement; consensus-seeking agents [7] converge on one shared value; AgenticPay [24] is price-only; CLiMRS [32] forms subgroups, not issue-bundled deals; a 2025 embodied-AI survey [12] does not contain negotiation, auction, or bargain in its accessible text. RoCo/CoELA-style dialogue for multi-robot planning (RoCo, ICRA 2024) is the closest structural neighbor — dialogue over sub-task plans and joint motion, not issue-bundled trades. At survey level, the CSUR MRTA review [2] lists no negotiation, bargaining, logrolling, or multi-issue technique (abstract and reference list checked via open mirror; full text paywalled).

What remains unclaimed. (and what this benchmark occupies): (i) a *physically grounded* system in which ≥ 2 robots strike one agreement bundling ≥ 2 *heterogeneous physical issue types* (lossy energy, cargo, claim rights) under executed dynamics; (ii) the Nash *bargaining solution* [26] (not Nash equilibrium) between robots; (iii) an empirical benchmark of such deals against auction and rule baselines. Honest limits: absence-of-evidence structure (an occupant could hide in an unindexed venue, non-English literature, or unqueried vocabulary — "multi-attribute contracting," "OCSM," "resource allocation by negotiation," "barter," "resource exchange protocol"); four load-bearing sources [2, 9, 17, 20] remain below full-text verification after a second pass (2026-07-23: all four paywalled; abstracts, the CSUR reference list, and an open-access 2026 survey in the same intersection contain no contradiction); the most plausible hidden occupant remains RoCo/CoELA-style dialogue making de facto multi-issue trades without negotiation vocabulary; Klein/Faratin complex-contract threads were not exhaustively checked for physically grounded instantiations. Current to mid-2026.

3 THE BENCHMARK

3.1 World

The base world (v2) is a 32×32 deterministic gridworld: two ore sources (16 and 40 Manhattan from the sink), one sink, one 2-slot charger dispensing 4 energy/tick, 120 ore units, a 2500-tick horizon, N=24 robots interacting within Chebyshev distance 2. Movement costs energy (more when loaded); inter-robot energy transfer loses 25%; each executed deal debits 0.05 battery per side. A robot is *stranded* at battery < 5 more than one cell from the charger. Later stages extend the world without changing the dynamics discipline: two companies with refineries and a refining tariff τ (v4); a rich ecology — 10 asteroids, 4 company-owned chargers with guest pricing, 240 units, lean fleets (v5); non-stationary contested fields with seeded arrivals and departures (v11–v12).

Heterogeneity is a **mean-preserving σ dial**: capacity $= 3 + \sigma U(-2, 2)$; efficiency $= 1 + \sigma U(-0.5, 0.5)$ clipped to $[0.5, 1.5]$; battery $= 60 + \sigma U(-40, 40)$ clipped to $[10, 100]$. Fleet means $(3, 1.0, 60)$ are σ-invariant, test-pinned. This repairs a v1 flaw: σ was confounded with poverty (fleet energy fell ~40% as σ rose), making "heterogeneity" claims partly scarcity claims.

3.2 The arm ladder

One mechanism is added per rung, so adjacent rungs differ by exactly one mechanism (Table 1, Figure 1).

Bundles combine three issues — energy, cargo, and sector/claim rights — in a discrete contract space ($7 \times 7 \times 2$ in the rich stage); Figure 2 shows one executed bundle. Bundle evaluation and execution share one code path; evaluated Φ == executed Φ is asserted on every deal. `auction_ssi` shares every discipline with the other rungs

Table 1: The arm ladder: one mechanism per rung.

arm	base	mechanism added
`null`	movement policy only	— (the zero point)
`rules`	null + trophallaxis	altruistic threshold rescue (Moonjaita-style [25])
`auction`	rules + cargo handoff	MURDOCH-style bilateral single-issue scalar reassignment [13]
`auction_ssi`	rules + broadcast SSI phase	sequential single-item broadcast auction, Koenig lineage [21]: multi-bidder competition, truthful $\Delta\Phi$ bids, single-issue items by construction (added at review; registered pre-run in the 2026-07-23 addendum)
`team`	null + greedy joint-Φ	cooperative ceiling: $\arg\max(\Phi_a + \Phi_b)$ over the same bundle space, no IR
`team[energy]`	team, energy only	strong Φ-informed single-issue baseline
`snhp`	null + Nash bundles	IR bargaining: Nash bargaining solution over bundles, strictly positive surplus both sides
`snhp+net`	snhp + trophallaxis fallback	isolates the removed-rescue confound

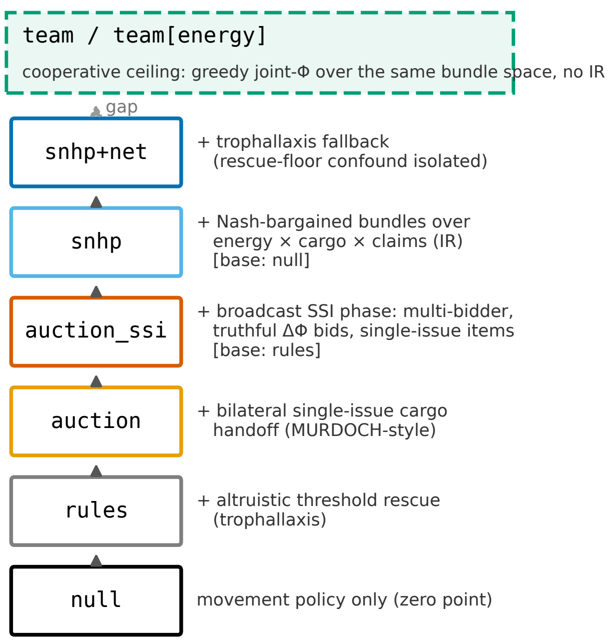

Figure 1: The arm ladder: one mechanism per rung, null → rules → auction → auction_ssi → snhp → snhp+net, with team / team[energy] as the cooperative-ceiling rail. `[base: x]` tags mark rungs whose base is not the previous rung (auction_ssi = rules + SSI; snhp = null + bundles).

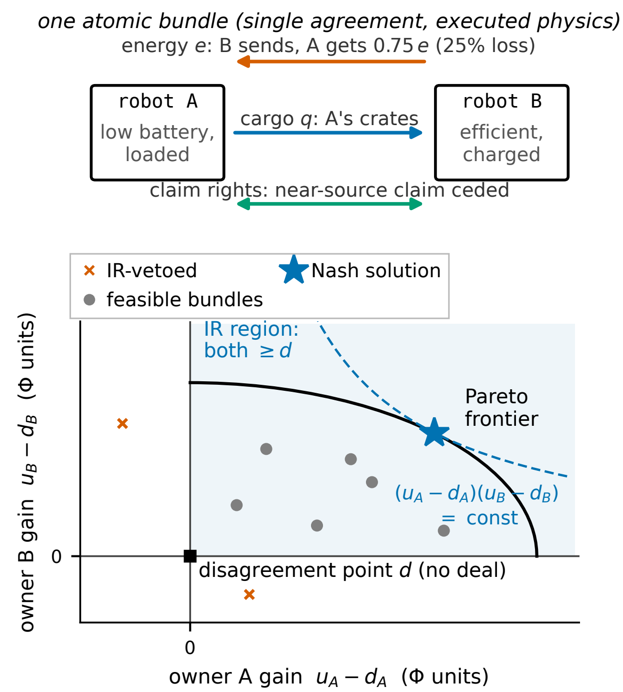

Figure 2: One executed bundle between two robots: the three issue axes (energy transfer with 25% loss, cargo, claim rights) and the utility-gain space with the IR region, disagreement point, Pareto frontier, and Nash-product hyperbola. Geometry illustrative; the loss rate and issue space are the spec constants of Sections 3.1–3.2.

(same item magnitudes, same 1.1 clearing hysteresis, same transaction cost, pause, cooldowns) and differs from `auction` by exactly {broadcast multi-bidder, energy-as-item, sequential rounds}; an item is exactly a $(q, 0, 0)$ or $(0, e, 0)$ bundle, so the arm structurally cannot logroll (asserted in code). Later rungs add hazard-priced risk (`-hz`), career-value pricing (`-lv`, `-lvc`), company walls, and a trust-gated joint tier (Section 4.5). The `team` arm exists because hostile review showed a trivial selfless control can dominate a bargaining arm; every bargaining claim faces both the auctions below and the ceiling above.

3.3 Metrics

Primary: **delivered ore at the fixed 2500-tick horizon** — never a ratio whose denominator shrinks when robots die (Section 3.4). Strandings are first-class: every headline is also reported as $\text{score}_k = \text{delivered} - k \cdot \text{stranded}$ at $k \in \{0, 2, 5\}$, after an audit showed the agents internally price a stranded drone at 1.5 ore while the scoreboard priced it at 0. Secondaries: energy efficiency metered to the last delivery, lost cargo, deal counts, multi-issue fraction, per-deal capture. Statistics: paired t and Wilcoxon on 24 seeds (16 in later columns), wins/n, Holm within families; makespan reported but not tested where horizon censoring dominates.

3.4 Registration protocol — and the v1 failure as method

The v1 headline claims failed hostile review: the efficiency metric *rewarded fleet death* (dead robots stop drawing charge; one run stranded all 24 robots, delivered less than the auction, and scored "a 5.8× win"); a trivial greedy joint-Φ control beat the bargaining arm on every metric at every σ; and the single-issue ablation struck zero deals — the failure that became C1. SPEC v2 was re-registered on 2026-07-14, *before any v2 results*, with claims C1–C3, the mean-preserving dial, the cooperative ceiling as a mandatory arm, delivered-at-horizon as primary, and explicit WIN/PARTIAL/KILL outcomes; every later extension (v3–v12, columns A–K, and review-response columns R1/GB/SSI) was registered pre-run with its own kill condition. A 2026-07-15 replay review then found two physics artifacts that 22 automated passes had missed — a cargo trap whose rescue subsidized our own mechanism by ~40% of the then-current bargaining-vs-auction gap, and temporally free deals; all columns were re-run, several verdicts shrank or died, and Section 4 reports post-correction numbers (detail, including a charger livelock, in Appendix A).

4 RESULTS

All verdicts are read against their pre-registrations. Sweep sizes: v2.1 888 runs; v3 960; v4.0 1944; v4.1 544; v5 736; v6–v7 480; columns G–K 16 seeds/cell; R1 (384 runs, 64 seeds), GB (360), SSI (216) per the 2026-07-23 addendum, whose headline test is Wilcoxon (p_w).

4.1 C1 — bundling is necessary under individual rationality

Registered prediction P1 (single-issue snhp arms strike 0 deals at every σ) **holds as amended**. Energy-only trade is structurally impossible: with 25% transfer loss and selfish utilities the donor always loses. Cargo-only trade exists solely as *distress jettison* (~0.5 deals/run): a near-stranded loaded robot gains by shedding load. Both are pinned as test invariants. In the full arms, ~98–99% of struck deals are multi-issue (per-run range 86–100%; HEAD-physics artifact `sweep_v2.1_head.json`). The cooperative form did **not** survive the physics correction: on the HEAD re-pin (R4), `team` beats `team[energy]` only at σ=0.75 (+1.62, p_w=.014; elsewhere −0.7 to +2.5, n.s.) — once deals cost time, extra bundle dimensions stop paying at the cooperative ceiling. The rebuttal of "single-issue suffices" rests on the structural IR half of C1 alone; the pre-correction "+1.8 to +17.1, significant at 4 of 5 σ" figure is retired to the correction history.

C1 recursed one level up: in the column-K information market (Section 4.7), a map-sync whose net value to the buyer is negative (bad news only) is IR-vetoed by construction; bad news trades only when bundled with enough good news to clear the veto. Single-issue bad news is unsellable; bundled truth trades.

4.2 C2 — sufficiency and the coordination gap

Every C2 comparison names its baseline: against the **bilateral MURDOCH-style handoff auction** (`auction`) bargaining wins delivered; against the **broadcast SSI auction** (`auction_ssi`) it does not — a registered kill (below, and Section 4.4).

All harsh-world numbers are the HEAD-physics re-pin (R4, registered 2026-07-23; artifact `sweep_v2.1_head.json`, 888 runs; the 2026-07-14 artifact is history only). `snhp+net` beats `auction` at every σ (+3.33 to +7.00 delivered; $p_w \le .014$ at $\sigma \in \{0, 0.25, 0.5\}$, p_w=.052 at σ=0.75, p_w=.011 at σ=1.0), with k2/k5 gaps larger and significant at every σ (σ=0.5 k5 +25.96, p_w=.0001) — the registered R4-C2 ordering, WIN (Figure 3). The v2.1-era ceiling inversion (119.6 vs 105.9 at σ=0) survives only in survival form: under HEAD the σ=0 delivered edge over `team` is +1.62 $[-0.27, +3.52]$ (p_w=.11 at 24 seeds; +1.77, p_w=.026 at the 64-seed R1a re-pin), carried by strandings (2.6 vs 4.1; k5 +9.58, p_w=.0002 at 64 seeds); the hive wins delivered outright at $\sigma \ge 0.75$ (−15.46/−12.00, $p_w \le .0002$). `snhp` beats `null` at every $\sigma \ge 0.25$ (+23.0 to +31.0, 23–24/24 seeds, $p_w \le .0001$) but **loses at** σ=0 (−6.29 $[-8.61, -3.97]$, p_w=.0001, 2/24): twin fleets have no gains from trade, so bargaining is pure overhead — as theory demands. Corrected honest negatives: at $\sigma \le 0.5$ pure `snhp` still never beats the auction on delivered (the low-σ win requires the rescue floor; at σ=0.75 it now does, +7.21, p_w=.0089); the pre-correction "net hurts at σ=0.75" tax (99.3 vs 90.4) was a physics artifact — under HEAD it vanishes (snhp − snhp+net = +1.58, n.s.) and the net helps significantly at every other σ.

The **coordination gap** (team − snhp on delivered; registered as "price of selfishness," renamed here) is real at every σ and grows past mid-σ: 12.0 → 4.2 → 5.2 → 13.9 → 16.7 as σ goes 0→1 (HEAD re-pin; all $p_w \le .048$). Registered prediction P3 (that it *shrinks* with σ) remains **refuted**: the gap dips at mid-σ, then grows. The gap is not the price of anarchy [22]: PoA bounds worst-case *equilibrium* welfare against the *optimum*, whereas ours is the measured difference between one IR mechanism and a greedy joint-utility heuristic — neither an equilibrium notion nor an optimum bound (the ceiling itself is non-optimal, Section 6). It is an empirical gap for this world and ladder, not a named theoretical quantity.

The physics correction shrank but did not kill the bilateral comparison on the rich stage: snhp-hz +3.4 (p=.005, was +9.6 — ~40% of the old gap was the pad subsidy), positive at every counterparty-noise level (+0.8..+4.1, $p \le .024$; the v5 noise dial found no crossover — the true-loss veto turns estimation error into failed proposals, not bad deals); on k-scores only the safety-net arm survives (snhp+net vs auction +4.5 delivered, +20.4 at k=5, both significant). The "market beats the hive" claim, re-pinned at 64 seeds (R1a, WIN): +1.77 delivered $[+0.32, +3.21]$, p_w=.026, strandings 2.27 vs 3.83, k5 +9.58, p_w=.0002; the seeds-0..15 subset reproduces the previously unversioned 16-seed numbers essentially exactly (+2.12, p_w=.041). On richer stages the hive wins at every k. The scoped law: *the safety-netted market beats central planning when survival is the binding constraint; the hive wins when logistics are.*

Two-geometry replication (R2/column GB: WIN).. On geometry B — sources at Manhattan 32/24 in opposite quadrants (no cheap 16-hop source, no 40-hop death-march), refinery mid-edge, mid-field

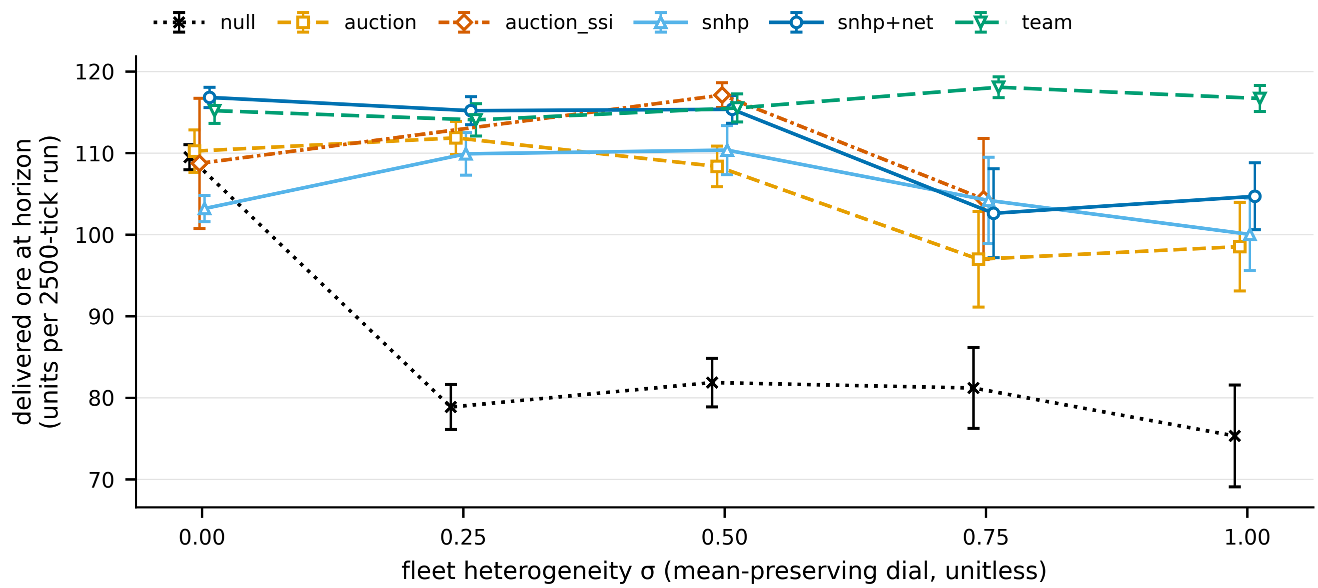

Figure 3: Delivered-at-horizon vs σ (mean ± 95% CI, 24 seeds/cell) for null, auction, auction_ssi, snhp, snhp+net, and team on the HEAD-physics v2.1 grid (`sweep_v2.1_head.json`); the auction_ssi series is from `sweep_v4_SSI.json` ($\sigma \in \{0, 0.5, 0.75\}$ only).

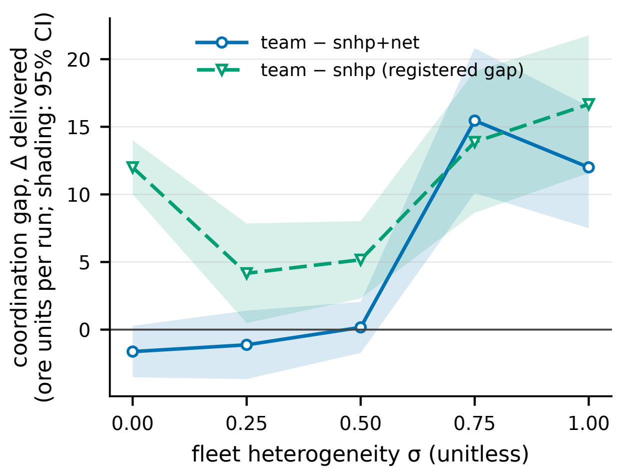

Figure 4: The coordination gap vs σ: paired Δ delivered with 95% CI bands for *both* definitions — team − snhp (the registered gap, formerly "price of selfishness"; this is the curve the text's 12.0 → 4.2 → 5.2 → 13.9 → 16.7 numbers refer to) and team − snhp+net (the survival-form comparison against the safety-netted arm). The two diverge sharply at $\sigma \le 0.5$, where the rescue floor does most of its work.

charger, non-mirrored — (snhp+net − auction) delivered is positive at every σ: +6.42 (p_w=.0002) at σ=0, +3.83 (p_w=.0038, 18/24) at σ=0.5, +4.42 (p_w=.0007, 19/24) at σ=1.0, with no k2 reversal (σ=0.5 k5 +26.96, p_w=.0001). The ladder ordering replicates descriptively. Nuance, reported as registered: on B the hive is *not* edged on raw delivered at σ=0 (118.8 vs 119.5, at the 120 ceiling) — but team strands 9.17 vs 0.96, so the net dominates every k-score; the survival form of the law holds on both maps. The bilateral comparison in C2 is a two-geometry result.

The strong-market test (R3/column SSI: KILL FIRED).. The registered prediction that snhp+net out-delivers SSI at $\sigma \ge 0.5$ died in its strong form: auction_ssi *significantly beats* the bargaining arm on delivered at σ=0.5 (−1.75 [−3.48, −0.02], p_w=.038; σ=0.75 −1.75, n.s.). The SSI market pays in strandings: snhp+net wins score$_{k=5}$ at σ=0.75 (+12.42, p_w=.027) and dominates the k-scores at the σ=0 sanity cell (k2 +19.92, k5 +37.67, both $p_w \le$.0001). The kill was earned against a real opponent: auction_ssi beats the bilateral auction by +8.75 (p_w=.0001) at σ=0.5 and +7.38 (p_w=.045) at σ=0.75. The honest statement of C2: *against a strong market baseline, bargaining buys survival, not throughput* — the fourth time a strengthened market matched or beat bargaining on raw output while losing on fleet survival (v3 hazard pricing, v9 career pricing, the v11 moving field, now SSI).

4.3 C3 — heterogeneity scaling

Registered prediction P4 (snhp − bilateral auction grows monotonically in σ) is **partial** and survives the HEAD re-pin (R4-C3, WIN): −7.0 → −2.0 → +2.0 → +7.2 → +1.5 delivered across $\sigma \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (`sweep_v2.1_head.json`) — point-estimate monotone over σ=0→0.75 (a 14.3-unit swing, ~60% smaller than the pre-correction 36.6; +7.21 at σ=0.75, p_w=.0089), saturating or breaking at σ=1.0. The break is unexplained (candidates: robots no deal can salvage; charger binding); we do not claim full monotonicity. The σ=0 end is now significantly negative (−7.04, p_w=.0001): twin fleets have nothing to trade, so bargaining is overhead on this world; in the two-company v4 world at σ=0 all mechanisms statistically tie `null` (post-correction) — the same no-heterogeneity-no-gains prediction at each world's cost structure. The strongest effect is the two-company v4 world: snhp beats auction +15.5/+15.0 delivered at σ=0.5/1.0 (p<0.001, 22–23/24 seeds) pre-correction; +14.1 (p<.001) on the corrected v4 preset. Geometry scoping: the gap's *sign* at σ≥0.5 replicates on geometry B (Section 4.2), but the five-point gradient is geometry-A only — on B the gap is already significant at σ=0 (+6.42, p_w=.0002; B's distances keep charging economics binding even for twin fleets), so the growth-with-σ shape remains a geometry-A result.

4.4 Killed claims (reported as registered)

The registered kill conditions fired repeatedly; the decision-relevant deaths are listed inline.

- **"Beats the market lineage" (R3/column SSI): KILL FIRED, the strong form.** The broadcast SSI baseline significantly out-delivers snhp+net at σ=0.5 (−1.75, p_w=.038) and never loses on delivered at σ≥0.5, while snhp+net keeps the survival-adjusted scores (k5 +12.42, p_w=.027 at σ=0.75). Registered consequence, applied without spin: every C2 delivered claim names its baseline — the bilateral MURDOCH-style handoff — and no general "beats the market lineage" language survives (numbers in Section 4.2). Structural note: auction_ssi is single-issue by construction and cooperative (truthful $\Delta\Phi$ bids, no payments); its awards are not IR (the losing side eats a negative $\Delta\Phi$ whenever 1.1 × |loss| < gain), so C1 — single-issue *IR* trade is infeasible — stands untouched.
- **Hazard-priced risk as a substitute for the rescue floor (v3): killed.** Both registered kill triggers fired (snhp-hz ≤ snhp at σ=0.25, −8.4, p=0.034; the net still added +22.8/+25.1 delivered at σ≤0.25). The surviving v3 "regime law" (risk pricing wins when risk is heterogeneous) then **died under corrected physics**: its crossing was largely the pad-subsidy artifact (best remaining edge +3.6, p=.17, with 15–21 strandings vs the net's 2–7).
- **Career-value drone pricing (v9/column H): kill fired.** Pricing drones at their remaining mining career made late-game drones disposable (delivered −16.5, p=.023; stranded +7.3, p=.001 vs flat pricing). With a capital floor it set the fastest selfish makespan recorded (688) and tied the net on delivered (239.6) — but still lost score$_{k=5}$ by −36 (p=.001). Three attempts to replace the rescue floor with a smarter market (v3, v6, v9) died on pre-registered criteria: *the safety net's edge is institutional, not a pricing bug.*
- **The density theory (v8/column G): killed as stated.** "Dense fields favor auction, sparse favor bargaining" was replaced by a hump: snhp+net − auction = +4.1 (grid 24) → +4.5 (32) → +7.3 (48) → −2.7 (64); see Figure 5. A market needs enough logistics friction to be worth paying for and enough meeting density to convene. Single-geometry; scoped accordingly.
- **"Cooperation is 43% faster": died under corrected physics.** Once deals cost time, the joint tier's speed edge is ~9% n.s.; the dividend moved to survival (gated fleets end 1.06 stranded vs the veto tier's 15.31).

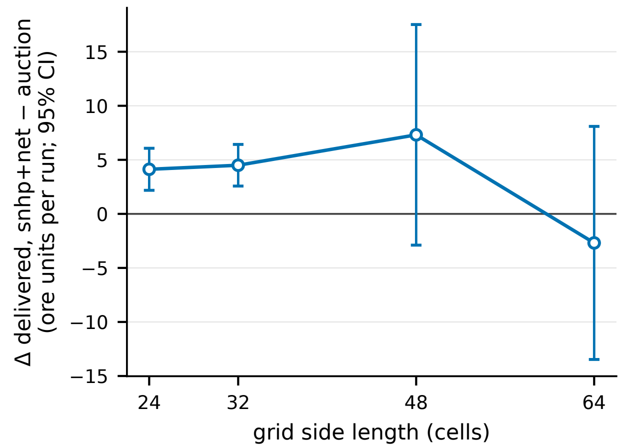

Figure 5: The v8 hump (single geometry; v5 preset, σ=0.5, 16 seeds/cell): (snhp+net − auction) delivered vs grid side 24/32/48/64, paired Δ with 95% CI whiskers (+4.12 / +4.50 / +7.31 / −2.69**).**

4.5 Deception and self-knowledge error: scoping note

The program's integrity results belong to a companion paper; here they are only scoped. The v6.0 registered kill fired informatively: Nash-IR bargaining with a true-loss veto is intrinsically deception-tolerant — lying barely pays because every executed deal clears both true disagreement points; exploitation requires the trusting joint tier, where attestation gating drops the liar advantage to statistical zero. v7 showed self-knowledge error (a miscalibrated gauge) leaves output flat while silently signing deals at negative true surplus. The relevance to C2: the bargaining tier's IR guarantee is exactly as good as each robot's knowledge of its own state, and information quality was a treated variable, not an assumption.

4.6 Column J — the moving field inverts a headline

On static fields, perfect field information was never load-bearing (column I: oracle − belief = +0.2 delivered, n.s.; the registered whole-column kill fired). Column J removed the crutches: seeded arrivals nobody knows about, departures that leave ghosts on stale maps, contested unmirrored ground.

- **P16a failed as registered**: oracle − belief on delivered is +11.4, directional but n.s. at 16 seeds (p=.15). The *significant* information-value channel is the books: belief-mode signs +7.0 more truly-harmful deals per run (p=.0004).
- **P16b inverted, significantly**: the auction, with substantially staler maps (358.4 vs 262.8 ticks at the 64-seed re-pin, p_w=.029; 16-seed: 305 vs 190), captures *more* of the newly arrived stock: +8.98 [+0.44, +17.52], p_w=.022 (42.7

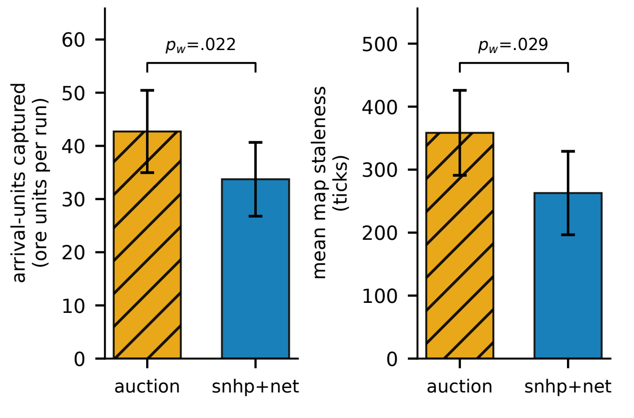

Figure 6: The column-J inversion at the 64-seed re-pin (R1c cells of `sweep_v4_R1.json`; moving + contested + belief-mode, no scouting or map market): arrival-units captured (42.70 vs 33.72, p_w=.022) and mean map staleness (358.4 vs 262.8 ticks, p_w=.029), auction vs snhp+net. The arm with fresher maps captures *less* of the new stock — freshness ≠ discovery.

vs 33.7 units/run) at 64 seeds (R1c, WIN); 16-seed original 46.2 vs 32.4, p=.03 — shrank from +13.8 but holds (Figure 6). Discovery needs physical coverage: the deal economy converges robots onto known-profitable loops, while the auction's inefficient wandering is accidental exploration.

- **Exploratory, flagged post-hoc**: on the moving contested field the auction out-delivers every coordination arm on raw gold (snhp+net −12.6, p=.044 at 16 seeds; +10.19, p_w=.011 at 64 seeds — reported but never registered, so it stays post-hoc) while k2/k5 remain a wash (+7.1 n.s. / +2.5 n.s.: the auction pays its gold edge in dead drones). The discovery gap explains the delivered gap almost exactly (13.8 arrival-units ≈ 12.6 delivered at 16 seeds). **Optimization buys blindness to novelty**: the bargaining advantage is a known-field phenomenon; novelty-rich worlds reward coverage over coordination.

4.7 Column K — scouting fixes discovery; the map market fixes the books

Column K priced the unknown. The discovery deficit died to *policy*, not markets: two scouts per company erased the auction's explorer edge (arrivals gap −13.8, p=.03 → −3.4, n.s.; delivered 284.1 vs 286.6) and made the oracle redundant (282.8 vs 284.1 — with patrols, information is free again). A Nash-priced map market (40 executed syncs/run) added nothing to discovery (−0.7, p=1.0) but **cut poisoned deals** ~30% — re-pinned at 64 seeds (R1b, WIN): **5.00 → 3.53 (−29%),** p_w=.0007 (16-seed original: 5.38 → 3.75, p=.04), with delivered a descriptive wash (+4.53, p_w=.46): *traded information's product is audit integrity, not output* (Figure 7). The registered bad-news trap (P17c) confirmed structurally: bad-news-only syncs are IR-vetoed; bundled truth trades (Section 4.1). Prospecting claims

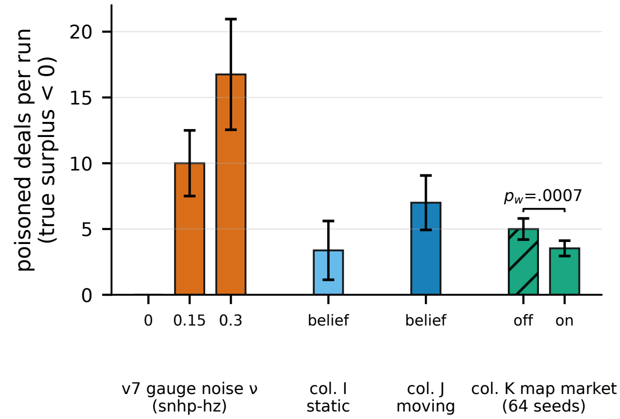

Figure 7: Poisoned deals per run (executed at negative true surplus for an honest party) across the program's error sources: v7 gauge noise ($\nu \in \{0, 0.15, 0.3\}$; snhp-hz, no margin, 32 seeds), column I static-field beliefs (16 seeds), column J moving-field beliefs (16 seeds), and column K map market off/on at the 64-seed re-pin (5.00 → 3.53, $\Delta = +1.47$ $[+0.69, +2.24]$, p_w=.0007). Output stays flat in every condition; only the books move.

produced measurable patrol differentiation (staleness 22.7 vs 25.4 inside vs outside own claims) with zero output effect at this scale.

5 DISCUSSION

Mechanism choice for cross-organization fleets is regime-dependent, and the regimes are now mapped. Bundling is necessary for any IR trade (C1, structural). Bargaining plus an unconditional rescue floor out-delivers a bilateral handoff auction on two geometries and, on survival-bound worlds, edges past the cooperative ceiling (C2); against the broadcast SSI lineage its edge is survival-priced, not throughput — continuous with the standing result that the safety net's value is institutional. The advantage needs heterogeneity (C3), mid-range logistics friction with enough encounter density (the v8 hump, single-geometry), and a *known* field (column J). Where those fail, simpler mechanisms win honestly: coverage beats coordination on novelty-rich ground, SSI competition matches bargained throughput, and central planning wins when logistics rather than survival bind. For an operator: choose by what binds — throughput (a strong auction suffices), fleet survival (bargaining + a rescue floor), or discovery (coverage/scouting policy).

The border result is the deployment-relevant one. Two firms bargaining at their boundary statistically match a full merger (team 118.6 vs twofirm 117.6, n.s., corrected physics); selfless cross-company transfers are net-harmful; posted-price infrastructure tolls collapse against a bargaining fleet (peak toll revenue −63%, 72.1 → 26.8). You do not need to merge fleets — you need a bargaining layer at the boundary, with IR keeping the border priced rather than closed or exploited.

The integrity thread is deliberately excluded. Across four error sources — lies (v6), gauge miscalibration (v7), stale maps (v10), a moving field (v11) — output stays flat while individual books silently bleed, and column K's map market heals books, not output. That finding and its settlement-infrastructure implications form the companion paper.

6 LIMITATIONS

This is a mechanism benchmark; hardware grounding is future work.

- **Scale**: N=24 (16–24 seeds per cell) is a multi-robot economy, not a swarm; by Şahin's criteria [28] no swarm claim is made or licensed.
- **Deterministic gridworld**: physical grounding here means world-dynamics-derived utilities, not hardware — there are no kinematics, collisions, sensor/actuation noise, or comms dropout, and noise-free simulation results are presumed non-transferable across the reality gap [18, 19] until shown otherwise.
- **Full-information utilities**: robots know their own Φ exactly (v7 perturbs one input); counterparty utilities are estimated only in v5+. Strategic information revelation and opponent learning are touched only through the registered lie and noise channels.
- **Deal-space abstraction**: energy transfer (lossy, minutes), cargo handoff (precision docking), and claim swaps (instant, free) have very different hardware time constants and failure modes; treating them as commensurable in one atomic bundle needs justification before any hardware claim.
- **Static-vs-moving scoping**: the headline bargaining advantages hold on static or slowly-changing fields; column J shows the ordering can invert under non-stationarity. Claims are scoped accordingly.
- **Statistical power and geometry**: the three previously fragile headline numbers were re-pinned at 64 seeds (R1, all WIN) and the C2 bilateral comparison replicates on a second geometry (R2); but 16-seed columns still leave directional results unresolved (P16a: +11.4, p=.15), the v8 hump and later-column findings remain single-geometry, and the C3 σ-gradient shape is geometry-A only. Exploratory findings are flagged and must not be cited as registered.
- **Mechanism instantiations**: the auction lineage has two rungs (bilateral MURDOCH-style and broadcast SSI [21]), but combinatorial variants are untested; the cooperative ceiling is greedy joint-Φ, not an optimal planner, so the coordination gap is measured against a heuristic, not an optimum.

7 REPRODUCIBILITY

The pipeline is deterministic and seed-pinned. Code and artifacts are released as a self-contained repository at `github.com/ryuxik/bundling-or-silence` (public at publication; Apache-2.0; the 174-test suite passes on a clean clone; source hashes in `PROVENANCE.md`). The world, arms, and runner are `world.py`, `arms.py`, `run.py`; every column is regenerated by `run.py --column` with column id A..K, R1, GB, or SSI, then `run.py --analyze` on the sweep JSON (v2.1/v3 via `run.py` directly; review-response numbers print from the committed `repin_report` path, not notebooks). Sweep artifacts (per-run rows with seeds, arm configs, event logs) are committed under `research/swarm/results/` (e.g. `sweep_v2.1.json`, 888 runs; `sweep_J.json` / `sweep_K.json`, 16 seeds/cell; `sweep_v4_R1.json`, 384 runs at 64 seeds; `sweep_v4_GB.json`, 360 runs; `sweep_v4_SSI.json`, 216 runs). The test suite (`pytest test_swarm.py`, 174 tests green as of the addendum) pins the structural claims (C1 zero-deal invariants, σ mean-preservation, evaluated==executed dynamics, attested-all ≡ honest-all bit-identity, the auction_ssi single-issue assertion) and regression-pins the corrected physics. Disclosure: the bargaining primitives (`generate_contract_space`, `filter_pareto_frontier`, `find_nash_bargaining_solution`) are the authors' deployed production negotiation engine's code, reused unmodified — deliberately, so the benchmark exercises the code path that ships. Pre-registrations, amendments, and fired kills are timestamped in `SPEC.md` and `SPEC-ADDENDUM-2026-07-23.md` (verdicts appended after runs; registrations never edited); the hostile v1 review, the standards brief, and the 2026-07-23 referee review are in `review/`.

A THE PHYSICS CORRECTIONS (DETAIL)

The 2026-07-15 replay review found: (1) a *pad-strand cargo trap* — a robot arriving at its target facility could strand on the arrival step and hold its cargo forever; rescue-capable arms ransomed that cargo back at 94–100% vs ~64% for auction/rules, a differential subsidy to our own mechanism worth ~40% of the then-current bargaining-vs-auction gap; and (2) *temporally free deals* — robots kept driving mid-exchange. Fixes: facilities unload on arrival (same tick); every executed exchange immobilizes both parties for DEAL_PAUSE=3 ticks. A third audit finding motivated the k-score policy: the scoreboard priced a dead drone at k=0 while the agents priced it at 1.5 ore internally, so every headline is reported at $k \in \{0, 2, 5\}$. Separately, a 22-agent pre-merge review found a *charger livelock* in the v7 column (the queue-release threshold read believed battery while the cap uses true battery, parking robots with gauge bias < -0.05 at chargers forever; reproduced 3×) plus a second unintended perturbation channel; both were fixed, regression-pinned, and columns D/E/F re-run, with pre-fix numbers struck in the results file.

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